Tuesday, October 27, 2026 2:45 PM

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LASSO — RIDGE

Regularisation via Bayesian inference.

LASSO/RIDGE are MAP solution under appropriate priors

Penalty

Can be used to provide information which may help us in other way (such as regularisation.

Variable selection in linear regression.

→ Many covariates are there but someof them have no effect ($\beta_j = 0$ for some j. such covariates can be more than one).

Classical setup: Fit model.

Based on p-value, you do variable selection.

(10) → (1) Forward selection | Backward selection → How to do in Bayesian setup.

$y = X\beta + \varepsilon$ which includes all covariates.

Then remove each of the covariate and check which has minm AIC

You compare full model and reduced model based on

M1 vs M2 : BF

$y = X\beta + \varepsilon$

Spike & Slab prior

$y \mid X, \beta, \sigma^2 \sim N(X\beta, \sigma^2 I_n)$

$\sigma^2 \sim IG(a_0, b_0)$

$\beta_j \mid \gamma_j = \begin{cases} 0 & \text{if } \gamma_j = \\ N(0, \sigma^2 \tau_1^2) & \text{if } \gamma_j \end{cases}$

$\gamma_j \sim Ber(\pi)$ $\forall j$ independently

$P(\gamma_j = 0 \mid X, Y)$

$\beta_j \mid \gamma_j, \sigma^2 \sim \gamma_j \delta_0 + (1-\gamma_j) N(0, \sigma^2 \tau_1^2)$ $\quad \beta_0 \sim N(0, \sigma^2 \tau_1^2)$

Gibbs sampling to estimate the parameters?

$\beta_0, \beta_1, \ldots \beta_p, \gamma_1, \ldots \gamma_p, \sigma^2$

Spike and slab prior for $\beta_1, \ldots \beta_p$, not for intercept.

$\beta_0^{(t+1)} \mid \beta_1^{(t)}, \beta_2^{(t)}, \ldots \gamma_1^{(t)}, \ldots \gamma_p^{(t)}, \sigma^{2(t)}$

$\beta_1^{(t+1)} \mid \beta_0^{(t+1)}, \beta_2^{(t)}, \ldots \gamma_1^{(t)}, \ldots \gamma_p^{(t)}, \sigma^{2(t)}$

$\gamma_1^{(t+1)} \mid \beta_0^{(t+1)}, \ldots \beta_p^{(t+1)}, \gamma_2^{(t)}, \ldots \gamma_p^{(t)}, \sigma^{2(t)}$

$(\beta_1, \beta_2, \gamma_1, \gamma_2, \sigma^2)$

$R^{(t+1)} \mid \beta_1^{(t)}, \gamma_1^{(t)} = 1, \gamma_2^{(t)}, \sigma^{2(t)}$

$\sigma^{2(t+1)} \mid \beta_2^{(t)}, \ldots, \beta_p^{(t)}, \gamma_1 \ldots \gamma_p^{(t)}$ |

$\Rightarrow \beta_1^{(t+1)} = 0$ w.p. 1.

$\gamma_1^{(t+1)} \mid \beta_2^{(t)}, \beta_1^{(t+1)}=0, \gamma_2^{(t)}, \sigma^{2(t)}$

$\gamma_1^{(t+1)} = 1$ w.p. 1

This violates irreducibility of the Markov chain & hence the sampler will not converge to the ~~the~~ target distn.

$\beta_1^{(t+1)} \mid \beta_2^{(t)}, \gamma_1^{(t)}=0, \gamma_2^{(t)}, \sigma^{2(t)}$.

Then $\beta_1^{(t+1)} \neq 0$ w.p. 1. $\quad \gamma_1^{(t+1)} \mid \beta_1^{(t+1)} \neq 0, \gamma_2^{(t)}, \sigma^{2(t)}, \beta_2 = 0$ w.p. 1

$\beta_j \mid \gamma_j \sim \gamma_j N(0, \tau_0^2) + (1-\gamma_j) N(0, \sigma^2 \tau_1^2)$ where τ_0 is very small. (Jiehering continuous spike & slab)

Gibbs sample (A, B, C) $\quad A \mid B, C$; $B \mid A, C$; $C \mid A, B$.

A, B, C need not be ~~scalar~~

$(\beta_1, \beta_2, \gamma_1, \gamma_2, \sigma^2)$

$\leftarrow (\beta_1, \gamma_1) \mid \beta_2, \gamma_2, \sigma^2)$

$(\beta_2, \gamma_2) \mid \beta_1, \gamma_1, \sigma^2)$

$\sigma^2 \mid \beta_1, \gamma_1, \gamma_2, \beta_2$

~~$(\gamma_1 \mid \beta_2, \gamma_2, \sigma^2)$~~ then $(\beta_1 \mid \gamma_1, \beta_2, \gamma_2, \sigma^2)$

Collapse Gibbs sampler $\Rightarrow$

$(\gamma_j, \beta_j) \mid \gamma_{-j}, \beta_{-j}, \sigma^2 \quad (\gamma_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$ & $(\beta_j \mid \gamma_j, \gamma_{-j}, \beta_{-j}, \sigma^2)$

$\gamma_j \in \{0, 1\}$ $\quad P(\gamma_j = 0 \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$ & $P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$

$\hat{p}_j$

$u \in (0, 1)$

$u \sim \text{Uniform}(0,1)$ if $u < \hat{p}_j$ then $\gamma_j = 0$

else $\gamma_j = 1$.

When $\gamma_j = 1 \Rightarrow \beta_j = 0$.

When $\gamma_j = 0$ then we generate β_j from $(\beta_j \mid \gamma_j, \beta_{-j}, \sigma^2)$

$P(\gamma_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$

$= \int P(\gamma_j, \beta_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \, d\beta_j$ ~~$= P(\gamma_j, \beta_j \mid \gamma_{-j}, \beta_{-j})$~~

$P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2)$

$P(\gamma, \beta \mid \gamma, \beta_{-j}) \propto P(\gamma_j) P(\beta_j \mid \gamma_j) P(y \mid \beta, \sigma^2)$

$$= P(\gamma_j)\, P(\beta_j \mid \gamma_j)\, \frac{1}{(\sigma\sqrt{2\pi})^n} \exp - \left\{ \sum_{i=1}^{n} \frac{\left(y_i - \underbrace{\sum_{k \neq j} x_{ik}\beta_k}_{z_i} - x_{ij}\beta_j \right)^2}{2\sigma^2} \right\}$$

$$P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \propto \pi \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{ - \frac{\sum z_i^2}{2\sigma^2} \right\}$$

Prior for γ_j is $\bar{\pi}$

$$P(\gamma_j = 0 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \propto \int (1-\bar{\pi}) \frac{1}{\sigma\tau_1\sqrt{2\pi}} \exp - \left\{ \frac{\beta_j^2}{2\sigma^2\tau_1^2} \right\} \frac{1}{(\sigma\sqrt{2\pi})^n} \exp - \left\{ \frac{\sum (z_i - x_{ij}\beta_j)^2}{2\sigma^2} \right\} d\beta_j$$

$$= (1-\bar{\pi})$$

$z_j = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}$

$z_j \mid \beta_j \sim N(x_j \beta_j, \sigma^2 I_n)$ — ①

$\beta_j \sim N(0, \sigma^2\tau_1^2) \Rightarrow x_j\beta_j \sim N(0, \sigma^2\tau_1^2 x_j x_j^T)$ — ②

$N(0, \sigma^2(I_n + \tau_1^2 x_j x_j^T))$

$= (1-\bar{\pi})\, f(z_i; \sigma^2(I_n + \tau_1^2 x_j x_j^T)) = a_j$

$P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) \propto \frac{\bar{\pi}}{(\sigma\sqrt{2\pi})^n} \exp\left\{ - \frac{z_i^2}{2\sigma^2} \right\} = b_j$

then, $P(\gamma_j = 1 \mid \gamma_{-j}, \beta_{-j}, \sigma^2) = \frac{b_j}{a_j + b_j} = p_j$

$\gamma_j \mid \gamma_{-j}, \beta_{-j}, \sigma^2 \sim \text{Ber}(p_j)$

if $u < p_j$, then $\beta_j = 0$

else $\left(\beta_j \mid z_j, \sigma^2, \gamma_j = 0 \right) \sim N(m_j, v_j)$ (γ|w)

where $m_j = \frac{\tau_j^2 x_j^T z_j}{1 + \tau^2 x_j^T x_j}$

$v_j = \frac{\sigma^2\tau^2}{1 + \tau^2 x_j^T x_j}$

$(\gamma_1, \beta_1 \mid \gamma_{-1}, \beta_{-1}, \sigma^2)$

$(\gamma_2, \beta_2 \mid \gamma_{-2}, \beta_{-2}, \sigma^2)$

$\vdots$

→ Updated

$(\sigma^2 \mid \gamma, \beta)$

$\beta_j \mid \gamma_j \underset{\sigma^2}{\sim} \gamma_j S_0 + (1-\gamma_j) N(0, \sigma^2\tau_1^2)$

$\pi(\sigma^2) \propto \frac{1}{\sigma^2}$ $\pi(\sigma^2 | \underline{\gamma}, \underline{\beta}) \propto \pi(\sigma^2) \prod_{j=1}^{p} \pi(\gamma_j) \pi(\beta_j | \gamma_j, \sigma^2) \frac{1}{(\sigma\sqrt{2\pi})^n} \exp\left\{ \frac{\|y - X\beta\|^2}{2\sigma^2} \right\}$

$\beta_j | \sigma^2 \sim N(0, \sigma^2 c^2)$

$\Downarrow$

$\pi(\sigma^2 | \beta) = IG\left(\frac{n+p}{2}, \frac{\|y - X\beta\|^2 + \sum \beta_j^2 / c^2}{2}\right)$

$\sigma^2 | \beta, \gamma, y \sim IG\left(\frac{n+k}{2}, \frac{\|y - X\beta\|^2 + \sum \beta_j^2 / c^2}{2}\right)$

where k = no. of γ's which are 0.

$P(\beta_j \neq 0 | \text{Data})$

$1 - \frac{\sum \gamma_j}{2000}$

10000

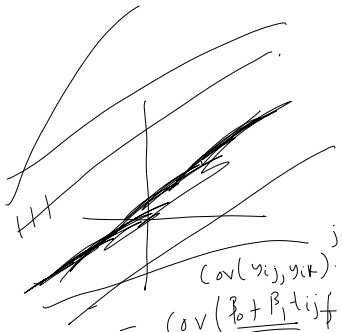
2000 ✓

$(\beta_j, \gamma_j, \sigma^2)$ - - - ($\forall j = 1(1)p$

Random-effects

$i = 1, 2, \ldots n$ (patients)

For each i, we have n_i observations



$= cov(\beta_0 + \beta_1 t_{ij} + b_i + \varepsilon_{ij}, \beta_0 + \beta_1 t_{ik} + b_i + \varepsilon_{ik})$

$= \tau^2$

$cor(y_{ij}, y_{ik}) = \frac{\tau^2}{\tau^2 + \sigma^2}$ & $cov(y_{ij}, y_{mn}) = 0$ if $i \neq m$

$y_{ij} = \underbrace{\beta_0 + \beta_1 t_{ij}}_{\text{Fixed effect}} + b_i + \varepsilon_{ij}$

$\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$

$b_i \overset{ind}{\sim} N(0, \tau^2)$

β_0: Population intercept

β_1: Population slope

b_i: random intercept for ith individual

σ^2: the within individual error variance

τ^2: Between-individual error variance

$(\beta_0, \beta_1, \sigma^2, \tau^2)$

Prior over the parameters

Δ then you try to find posterior

$cov(y_i, y_j) = 0$

$cov(y_{ij}, y_{ik}) = \tau^2$ $j \neq k$

$V(y_{ij})$ --